

MATH 344: Exam 2
Spring —₁ 2026
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50 Minutes

Name: Caleb McWhorter — Solutions

Write your name on the appropriate line on the exam cover sheet. This exam contains 7 pages (including this cover page) and 6 questions. Check that you have every page of the exam. Answer the questions in the spaces provided on the question sheets. Be sure to answer every part of each question and show all your work. If you run out of room for an answer, continue on the back of the page — being sure to indicate the problem number.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
Total:	60	

1. (10 points) Let $\mathbf{b} = \begin{pmatrix} 3 \\ -4 \\ 11 \end{pmatrix}$ and A be the matrix with LU factorization

$$A = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 4 & -3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix}$$

- (a) Compute $\det A$.

We use the fact that $\det(MN) = \det M \cdot \det N$ and the determinant of a triangular matrix is the product of the diagonal entries. Therefore, we have...

$$\det A = \det(LM) = \det L \cdot \det M = 1 \cdot (1 \cdot 3 \cdot 4) = 12$$

- (b) Use the LU factorization above to solve the matrix-vector equation $A\mathbf{x} = \mathbf{b}$.

We know that $A = LU$, so $\mathbf{b} = A\mathbf{x} = (LU)\mathbf{x} = L(U\mathbf{x})$. We also know that $U\mathbf{x}$ is a vector, call this vector $\mathbf{y}$. So, we have $L\mathbf{y} = \mathbf{b}$. Therefore, we have...

$$\begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 4 & -3 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -4 \\ 11 \end{pmatrix}$$

We solve this system using forward-substitution:

$$\begin{aligned} 1y_1 + 0y_2 + 0y_3 &= 3 &\implies y_1 &= 3 \\ -1y_1 + 1y_2 + 0y_3 &= -4 &\implies -3 + y_2 &= -4 &\implies y_2 &= -1 \\ 4y_1 - 3y_2 + 1y_3 &= 11 &\implies 4(3) - 3(-1) + y_3 &= 11 &\implies y_3 &= -4 \end{aligned}$$

Because $U\mathbf{x} = \mathbf{y}$ and $\mathbf{y} = \begin{pmatrix} 3 \\ -1 \\ -4 \end{pmatrix}$, we know...

$$\begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ -4 \end{pmatrix}$$

We can solve this using back-substitution:

$$\begin{aligned} 0x_1 + 0x_2 + 4x_3 &= -4 &\implies 4x_3 &= -4 &\implies x_3 &= -1 \\ 0x_1 + 3x_2 + x_3 &= -1 &\implies 3x_2 - 1 &= -1 &\implies x_2 &= 0 \\ 1x_1 + 2x_2 - x_3 &= 3 &\implies x_1 + 2(0) - (-1) &= 3 &\implies x_1 &= 2 \end{aligned}$$

Therefore, the solution is $\mathbf{x} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$.

2. (10 points) Consider the invertible matrix $A = \begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & 5 \\ -3 & -1 & 1 \end{pmatrix}$.

- (a) Explain why A does not have an LU decomposition without computing a PLU decomposition for A , i.e. do not use (b).

Because A is a square invertible matrix, A has an LU if and only if all the leading principal minors are nonzero. The first leading principal minor is $M_1 = \det(1) = 1 \neq 0$. However, $M_2 = \det \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix} = 1(6) - 2(3) = 6 - 6 = 0$. Therefore, A does not have an LU decomposition. But A must have a $PA = LU$ or $A = PLU$ decomposition.

- (b) Find a PLU decomposition for the matrix A .

Row Operations	Current L	Current U
$-2R_1 + R_2 \rightarrow R_2$ $3R_1 + R_3 \rightarrow R_3$	$\begin{pmatrix} 1 & 0 & 0 \\ ? & 1 & 0 \\ ? & ? & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 3 & -2 \\ 2 & 6 & 5 \\ -3 & -1 & 1 \end{pmatrix}$
$R_2 \longleftrightarrow R_3$	$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & ? & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 3 & -2 \\ 0 & 0 & 9 \\ 0 & 8 & -5 \end{pmatrix}$
—	$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}}_L$	$\underbrace{\begin{pmatrix} 1 & 3 & -2 \\ 0 & 8 & -5 \\ 0 & 0 & 9 \end{pmatrix}}_U$

We had to swap R_2 and R_3 . The permutation that only swaps R_2 and R_3 for a 3×3 matrix is $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. We then know $PA = LU$. So, $A = P^{-1}LU$. But because

P is a permutation matrix, we know that $P^{-1} = P^T$. So, $A = P^T LU$. We have $P^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. Therefore, we have...

$$A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}}_P \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}}_L \underbrace{\begin{pmatrix} 1 & 3 & -2 \\ 0 & 8 & -5 \\ 0 & 0 & 9 \end{pmatrix}}_U$$

3. (10 points) A matrix A and its reduced-row echelon form, R , are given below.

$$A = \begin{pmatrix} 1 & -2 & 0 & 0 & 3 \\ 3 & -6 & 1 & 2 & -1 \\ 2 & -4 & 1 & 0 & 6 \end{pmatrix}, \quad R = \begin{pmatrix} \boxed{1} & -2 & 0 & 0 & 3 \\ 0 & 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} & -5 \end{pmatrix}$$

(a) Determine which $\mathbb{R}^n$ each of the spaces below are a subspace of and find their dimension.

	Col(A)	Row(A)	Null(A)
Subspace of...	$\mathbb{R}^3$	$\mathbb{R}^5$	$\mathbb{R}^5$
Dimension	rank $A = 3$	rank $A = 3$	nullity $A = 2$

(b) Find a basis for Col(A).

We choose the pivot columns of A .

$$\left\{ \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \right\}$$

(c) Find a basis for Row(A).

We choose the nonzero rows of $\text{rref}(A)$.

$$\left\{ \begin{pmatrix} 1 \\ -2 \\ 0 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ -5 \end{pmatrix} \right\}$$

(d) Find a basis for Null(A).

We know Null(A) is a subspace of $\mathbb{R}^5$, so we have five variables: $x_1, \dots, x_5$. The basic variables are x_1, x_3, x_4 and the free variables are x_2, x_5 . We know $x_1 - 2x_2 + 3x_5 = 0$, $x_3 = 0$, and $x_4 - 5x_5 = 0$, i.e. $x_1 = 2x_2 - 3x_5$, $x_3 = 0$, and $x_4 = 5x_5$. But then...

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 2x_2 - 3x_5 \\ x_2 \\ 0 \\ 5x_5 \\ x_5 \end{pmatrix} = x_2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} -3 \\ 0 \\ 0 \\ 5 \\ 1 \end{pmatrix}$$

Therefore, a basis for the nullspace is...

$$\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 0 \\ 5 \\ 1 \end{pmatrix} \right\}$$

4. (10 points) Complete the parts given below.

- (a) Compute the following *without* explicitly computing the determinant: $\det \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -5 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$.

This is an elementary matrix that when multiplied on the left performs $-5R_1 + R_3 \rightarrow R_3$, which leaves the determinant unchanged. Therefore, we have...

$$\det \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -5 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = 1$$

- (b) Find an elementary matrix that scales the second row of a 2×2 matrix by $\frac{1}{3}$.

$$\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$$

- (c) If $E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ -2 & 0 & 1 \end{pmatrix}$ and A is a 3×3 matrix, describe the matrix EA .

The product EA has the same first row as A , the second row is 3 times the second row of A , and the third row is the third row of A replaced by twice the first row of A subtracted from the third row of A . That is, EA is A after the elementary row operations $3R_2 \rightarrow R_2$ and $-2R_1 + R_3 \rightarrow R_3$.

- (d) Compute the following *without* explicitly multiplying the matrices:

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 11 & -15 & 18 & -20 \\ 8 & 5 & 16 & -9 \\ 0 & -3 & 10 & 1 \\ 3 & 17 & 0 & 4 \end{pmatrix}$$

The matrix on the left is a permutation matrix which swaps rows one and three, and swaps rows two and four, i.e. $R_1 \longleftrightarrow R_3$ and $R_2 \longleftrightarrow R_4$. Therefore, we have...

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 11 & -15 & 18 & -20 \\ 8 & 5 & 16 & -9 \\ 0 & -3 & 10 & 1 \\ 3 & 17 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 0 & -3 & 10 & 1 \\ 3 & 17 & 0 & 4 \\ 11 & -15 & 18 & -20 \\ 8 & 5 & 16 & -9 \end{pmatrix}$$

5. (10 points) Complete the following parts:

(a) Is the set $\left\{ \begin{pmatrix} 1 \\ -2 \\ 0 \\ 5 \end{pmatrix}, \begin{pmatrix} 6 \\ -1 \\ 4 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ -5 \\ 0 \\ 1 \end{pmatrix} \right\}$ linearly independent? Explain.

No, the set contains the zero vector, $\mathbf{0}$. No collection of vectors containing the zero vector can be linearly independent.

(b) Can a set of 13 vectors in $\mathbb{R}^5$ be linearly independent? Explain.

No, the vectors cannot be linearly independent. The span of the set is clearly a subspace of $\mathbb{R}^5$. The only possible dimensions of a subspace of $\mathbb{R}^5$ are $0, 1, \dots, 5$. If the given vectors were linearly independent, then they would form a basis for the subspace generated by their span. This subspace would then have dimension 13, which is not possible. Therefore, the vectors must be linearly dependent. Alternatively, we know a subspace of dimension D cannot have a set of $N > D$ linearly independent vectors.

(c) Find a basis for the subspace $W = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}$ and find its dimension.

We know that $x + 2y - z = 0$ if and only if $z = x + 2y$. But then...

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ x + 2y \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

Therefore, $W = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right\}$. But these two vectors are not parallel. So,

this set is linearly independent. But then the set is a basis for W . This means that $\dim W = 2$. Indeed, $x + 2y - z = 0$ is the equation of a plane in $\mathbb{R}^3$ passing through the origin.

6. (10 points) Mark the following statements True (T) or False (F). You *do not* need to justify your answer.

(a) F : If $\text{nullity}(A) = 0$, i.e. $\dim \text{null } A = \{0\}$, then $A\mathbf{x} = \mathbf{0}$ does not have a solution.

(b) F : The number of independent rows and independent columns of a matrix can be different.

(c) T : A 12×9 matrix A can have rank at most 9.

(d) T : The columns of *any* 6×7 matrix are linearly dependent.

(e) T : If A is a 3×4 matrix with rank 3, then $A\mathbf{x} = \mathbf{b}$ *always* has a solution but the solution is *never* unique.